

C_AIG

SAP Certified - SAP Generative AI Developer

SAP CERTIFICATION TASKS

Course Version: 2604

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Typographic Conventions

American English is the standard used in this handbook.

The following typographic conventions are also used.

This information is displayed in the instructor's presentation	
Demonstration	
Procedure	
Warning or Caution	
Hint	
Related or Additional Information	
Facilitated Discussion	
User interface control	<i>Example text</i>
Window title	<i>Example text</i>

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Instructions

In this system-based assessment, you will be asked to complete several tasks in an SAP system.

Data Entry Accuracy

Some tasks will require you to enter your group number and other data values. **It is critical that these values are entered exactly as specified** or you risk losing points. Pay close attention to:

- Correct capitalization of letters
- Empty spaces and all other symbols must be maintained exactly as stated.
- When entering your group number, replace redundant hashtags with leading zeros.

Example:

Your group number is 2

Input Value	Correct	Incorrect
Admin_##	Admin_02	admin_02, Admin 02
New Lead-####	New Lead-0002	New Lead-2, New Lead0002

Please make a note of your group number when you enter the assessment system shortly.

SAP Joule for Consultants

During a system-based assessment, you can use open-book resources, including SAP Joule for Consultants. To access it:

- Select SAP Joule for Consultants from the systems menu, along with other system options, before starting the assessment.
- It opens in a separate browser window at no cost.
- SAP Joule for Consultants remains available for the entire duration of your assessment.

For all other questions, please refer to [FAQs](#).

When you have finished reading these task instructions and are ready to begin the timed assessment, please return to the browser where you downloaded this document and follow the instructions there.

Good luck!

Context

Scenario

A large online retailer struggles to manually process and analyze thousands of daily customer emails about various issues like product defects, delivery delays, and return requests. This manual approach is inefficient, inconsistent, and results in customer dissatisfaction.

Your task is to design, develop, and iteratively refine an AI solution using the SAP's generative AI hub. The solution should extract essential insights for fast routing, proactive problem-solving, and ongoing service enhancement.

This scenario consists of four parts. The estimated duration to finish the parts in minutes is added in parentheses. We recommend that you complete the parts in the following sequence.

1. Designing an Initial Prompt and Structuring for Core Feedback Insights (25m)
2. Refining Accuracy and Adding Semantic Depth for Problem Resolution (20m)
3. Evaluating and Selecting the Optimal LLM for Production (20m)
4. Creating an Advanced Multi-Issue Extraction & Prioritization Template (25m)

Deliverables

Your submission will consist of specific, versioned prompt templates, configured orchestration workflows, and traceable outputs demonstrating your problem-solving process and implementation. You will also be required to save specific AI responses directly into the **Assistant role** of your prompt template for tracking your solution's evolution. Save all your templates in the **orchestration** scenario.

Sample message for testing your prompts

Sample messages are provided at the end of this document. You can use these messages to develop your prompts as suggested in the tasks below.

Designing an Initial Prompt and Structuring for Core Feedback Insights (Estimated Time: 25 minutes)

Objective

Develop a foundational prompt template that accurately extracts essential customer feedback details into a structured JSON format, ready for immediate departmental routing.

Task 1.1: Design and Implement the Base Prompt (Prompt Template Version 1.0.0)

The customer service, product, and logistics teams need a quick, standardized way to understand the core of a customer's message. Create a new prompt template named **feedback-analyzer-CERT####, version 1.0.0** in SAP's generative AI hub to identify the main issue, sentiment, and any referenced product or order details.

1. Define appropriate **System** and **User** roles.
2. The **System** role must clearly state the LLM's primary task (customer feedback analysis) and the desired output schema.
3. The **User** role should contain instructions for processing the input message.
4. Add a variable **user_email_placeholder** where you will provide the input message.
5. The output **MUST** be a JSON object containing the following keys:
 - a. **Feedback_ID**: string (e.g., AUTO-FEED-001)
 - b. **Main_Category**: enum (e.g., Product Issue, Delivery Problem, Return/Refund, General Inquiry, Website/App Issue, Account Issue)
 - c. **Customer_Sentiment**: enum (Very Negative, Negative, Neutral, Positive, Very Positive)
 - d. **Issue_Summary**: string (concise summary, max 100 words)
 - e. **Order_Reference_ID**: string (e.g., #GGX12345, N/A if not present)
 - f. **Product_Reference**: string (e.g., "Smartwatch Model X", N/A if not present)
 - g. **Suggested_Department**: enum (Customer Service, Product Team, Logistics, Returns Processing, Technical Support)
6. Make sure the JSON response contains JSON output only with no additional text or formatting.
7. To develop an effective prompt template:
 - a. You can use the prompt editor and sample messages (located at the end of the document) to create and refine multiple versions of the prompts. You can save these prompts for future reference.

- b. Finally, select the most suitable version for use as the template, **feedback-analyzer-CERT#### version 1.0.0**.



Note:

You can use the sample messages provided in the Sample Messages section after the practice test description



Note:

Click on the top right corner in SAP AI Launchpad to find your unique user id in the form of "CERT####".

Example: If your group number is "0123 SAP Learning " your user id will be "CERT0123".

Your template name must include the exact words with your user ID. For example: **feedback-analyzer-CERT0123, version 1.0.0**, and the variable name **user_email_placeholder** for this task. Similarly, ensure correct versions for each of the following tasks like **1.1.0, 1.2.0, 2.0.0** etc. as mentioned in the task. Following these instructions correctly is critical for you to pass the certification.

Ensure that you save the prompt template with proper name and version number.

Expected outcome: A new prompt template with the name **feedback-analyzer-CERT####, version 1.0.0** with appropriate prompts in the system and user roles and a variable based on the above instructions.

Task 1.2: Implement Enterprise-Grade Prompt Hardening (Prompt Template Version 1.1.0)

Update your **feedback-analyzer-CERT####** template to **version 1.1.0** to implement prompt hardening by adding an "IMPORTANT" section within the **System** role.

This section must contain **explicit negative instructions** that strictly control LLM's behavior by forbidding actions that would:

- Compromise its **dedicated focus** on customer feedback analysis.
- Lead to the **unnecessary disclosure of any customer's personal data**, beyond the Order_Reference_ID or Product_Reference fields explicitly required in the JSON output.
- Result in a response that is not **clean, parseable JSON**, or unsuitable for automated systems.

Expected outcome: The updated prompt template with **similar** name **feedback-analyzer-CERT####**, and a new **version 1.1.0** with an additional section for prompt hardening based on the above instructions.

Task 1.3: Capture the Baseline Output for Evolution (Prompt Template Version 1.2.0)

Run your **feedback-analyzer-CERT####** template (**version 1.1.0**) in the Prompt Editor using the following sample customer email in the **user_email_placeholder** variable.

**Hint:**

You can use any model of your choice while executing your template in this task and all other remaining tasks. In case a particular model is not working, select another model and continue.

Once the JSON response is generated, **copy the JSON output and paste it into the Assistant role** of your template and save it to **version 1.2.0**. This will serve as a baseline to track how your solution improves.

**Note:**

When copying the response for this and all subsequent tasks, you can either select the text manually or use the copy button in the interface. Both methods provide the JSON format required for the next steps.

Subject: My recent order #GGX56789 - missing item

Hi,
I received my order #GGX56789 today, but the 'Wireless Headphones' listed on the invoice were not in the box. I am very frustrated as I needed them for a trip this weekend. Please send the missing item or issue a refund immediately. This is unacceptable.

**Note:**

In case you are unable to execute the prompt due to a greyed out **Run** button, click all the input fields (system, user, and variable fields) and check again. You can also try using another model to execute the prompt.

Expected outcome: The updated prompt template with a **similar** name **feedback-analyzer-CERT####**, and a **new version 1.2.0** with an additional **Assistant** role, based on the above instructions.

Refining Accuracy and Adding Semantic Depth for Problem Resolution (Estimated Time: 20 minutes)

Objective

Improve the LLM's ability to accurately categorize complex feedback and extract specific, actionable details crucial for resolution, not just categorization.

Task 2.1: Address Ambiguous Categorization (Prompt Template Version 2.0.0)

While knowing the Main_Category is good, service agents still have to read the full summary to understand what the customer *wants* (e.g., a replacement, a call, a technical fix). You need to extract specific resolution requests or critical details directly.

Update your **feedback-analyzer-CERT####** template to **version 2.0.0**. Modify the output schema to include a new field: **Actionable_Resolution_Points** (an array of strings) to extract 1-3 distinct, concise points detailing what the customer explicitly requests or what specific technical detail needs addressing for resolution.

Expected Outcome: The updated prompt template with a **similar** name **feedback-analyzer-CERT####**, and a **new version 2.0.0** with the additional new **Actionable_Resolution_Points** field based on the above instructions.

Task 2.2: Extract Actionable Resolution Details (Prompt Template Version 2.1.0)

Instructions: Run your **feedback-analyzer-CERT####** template (**version 2.0.0**) in the Prompt Editor using the following sample feedback. Once the JSON response is generated, **copy and save this specific JSON output into the Assistant role** of this template version. Save it to a new template **version 2.1.0**. This validates your refinement.



Caution:

Recall that to use the prompt template in Prompt Editor and generate fresh responses, you need to **delete the all the previous Assistant roles**.

Subject: Glitch on checkout page, Order #GGX98765 stuck
Hi Support,
I was trying to complete my purchase for Order #GGX98765 (items: "Portable Speaker" and "Travel Adapter"), but the website's checkout page crashed after I entered my payment details. Now the order is stuck, and I can't try again. I need someone to clear the pending order and allow me to repurchase, or help me complete it.

Expected outcome: The updated prompt template with a **similar** name **feedback-analyzer-CERT####**, and a new **version 2.1.0** with an additional **Assistant** role, based on the above instructions.

Evaluating and Selecting the Optimal LLM for Production (Estimated Time: 20 mins)

Objective: Systematically evaluate different LLMs using your fully refined prompt template. Based on this evaluation, you can configure your solution to use the optimal model for production deployment.

Task 3.1: Perform Comparative Model Evaluation and Selection

Using your template (**feedback-analyzer-CERT####** version 2.0.0), run the following **sample customer feedback messages** three times each time using a **different LLM model** available in the generative AI hub (e.g., GPT-4o, Mistral AI, Gemini)



Note:

In case a particular model is not working or is not available, select another model and continue. Your result will not be impacted by unavailability of a particular model.

Feedback Message (Delivery Problem - Neutral/Negative):

Subject: Delivery confusion for order #GGX33445

My order #GGX33445 (a "Premium Coffee Maker") was marked as delivered two days ago, but I haven't received it. My neighbor also didn't get it. The tracking link just says "delivered." Can you please investigate with the shipping company and give me an update?

1. After executing your prompt with each model, carefully review the generated JSON output for accuracy, completeness, and adherence to the specified format to ensure the most accurate customer feedback analysis. Pay close attention to Main_Category, Customer_Sentiment, Suggested_Department, and Actionable_Resolution_Points.
2. To do this, run your template with each model for the given Feedback Message and review the JSON output. Repeat the same for the next model.
3. After comparing the results decide which LLM performs best with your current setup.
4. Copy the JSON output for your selected model using **Feedback Message** and save it to the **Assistant** role of your **feedback-analyzer-CERT####** template to new **version 2.2.0**. Add the **"Optimal Model Output"** text on the top of the saved response. This text is important to track your results.

Expected outcome:

The updated prompt template with a **similar** name **feedback-analyzer-CERT####**, and a new **version 2.2.0** with an additional **Assistant** role based on the above instructions.

Creating an Advanced Multi-Issue Extraction & Prioritization Template (Estimated Time: 25 minutes)

Objective

Enhance the solution to handle multiple distinct issues within a single customer feedback message, requiring a more complex JSON structure. Introduce a "Priority" field for each identified issue.

Task 4.1: Implement Multi-Issue Support & Prioritization (Prompt Template Version 3.0.0)

Customers often voice several distinct complaints or questions within a single communication, for example a product defect, a shipping delay, and a billing question all in one email. The current schema is designed for a single Main_Category per feedback. You need to identify all distinct issues and also assign a priority to each to help agents triage effectively.

Update your **feedback-analyzer-CERT####**, template version 2.0.0 to a new version, **3.0.0**.

- Modify the output JSON schema significantly. The JSON will now comprise two levels, a header level and a line item level. The header level will consist of the fields Feedback_ID and Overall_Customer_Sentiment (representing the general tone of the entire message).
- The line item level should now be structured as an array where each element in the array is a separate JSON object representing a distinct customer issue. Each Issue element MUST contain:
 - **Issue_ID**: string (e.g., "ISSUE-A", "ISSUE-B") - a unique identifier for each distinct issue found within the current feedback.
 - **Main_Category**: enum (same as before: Product Issue, Delivery Problem, Return/Refund, General Inquiry, Website/App Issue, Account Issue)
 - **Customer_Sentiment_for_Issue**: enum (Very Negative, Negative, Neutral, Positive, Very Positive) - *specific to this particular issue*.
 - **Issue_Summary**: string (concise summary for *this specific issue*, max 75 words)
 - **Order_Reference_ID**: string (e.g., #GGX12345, N/A if not present, specific to this issue)
 - **Product_Reference**: string (e.g., "Smartwatch Model X", N/A if not present, specific to this issue)
 - **Actionable_Resolution_Points**: array of strings (1-3 distinct, concise points detailing what is needed for *this issue's* resolution).
 - **Priority**: enum (High, Medium, Low) - Based on the urgency, impact on the customer, and severity of *this specific issue*.

Expected outcome:

The updated prompt template with a similar name **feedback-analyzer-CERT####**, and a new **version 3.0.0** with updated prompts for the new JSON format in the **System** and **User** role.

Task 4.2: Capture Complex Multi-Issue Output for Validation (Prompt Template Version 3.1.0)

Run your **feedback-analyzer-CERT####** template (**version 3.0.0**) in the Prompt Editor using the following complex sample feedback. Once the JSON response is generated, copy and save this specific JSON output into the **Assistant** role of this **template version 3.1.0**. This validates your ability to handle multi-issue extraction.

Subject: Problems with my recent order and account!

Hello,
I'm writing because I have two serious issues. First, the "Smart Coffee Maker" (Order #GGX88990) I bought is defective; it keeps leaking. I want a replacement or a full refund. Second, my online account has been showing incorrect order history for months, and I can't update my shipping address. This is very frustrating, and I've tried calling customer service but couldn't get through. Please fix my account and send instructions for the coffee maker return.

Expected outcome:

The updated prompt template with a **similar** name **feedback-analyzer-CERT####**, and a new **version 3.1.0** with an additional **Assistant** role, based on the above instructions.

Sample Messages**Sample Message 1: Product Issue (Defective Product)**

Subject: Faulty "Smart Blender" - Order #BLEND1234

Hi Support,
I received my 'Smart Blender Pro X' (order #BLEND1234) two days ago, and it's already not working correctly. The motor sounds very weak, and it keeps stopping mid-blend. It's clearly defective. I would like a replacement unit sent out immediately, or a full refund if a replacement isn't possible. This is very frustrating as I need it daily for my morning shakes.

Sample Message 2: Delivery Problem (Lost Package)

Subject: Where is my order #PKG5678? Tracking shows delivered but I have nothing!

To whom it may concern,
My order #PKG5678 for the "Premium Coffee Subscription" was marked as delivered yesterday, but I haven't received anything. I've checked with my neighbors and the building management, and no one has seen it. This is a recurring issue with your delivery service! Please investigate with the shipping company and locate my package, or send a new one. I'm very upset about this.

Sample Message 3: Website/App Issue (Payment Error)

Subject: Cannot complete purchase - Payment Error on website

Hello,
I'm trying to buy the 'Ergonomic Office Chair' and some accessories, but every time I try to complete my payment on your website, it shows a generic 'payment error' message. My card is definitely working on other sites. I've tried multiple times, and now my cart contents might

expire! Can you please check what's wrong with your payment gateway? I need to complete this order.

Sample Message 4: General Inquiry (Pre-Sales Question)

Subject: Question about "Eco-Friendly Tablet Case" compatibility

Dear Team,
I'm looking at your 'Eco-Friendly Tablet Case' on your website. I have an older tablet model, the "TabPro 7" (from 2021). Will this case be compatible with it, or is it only for newer models? I want to make sure before I make a purchase. Thanks for your help!

Sample Message 5: Multi-Issue (Defective Product & Account Billing)

Subject: Two problems: broken headset & duplicate charge for membership!

I have two urgent issues to report! First, the 'Gaming Headset' (order #GAME9012) I bought last week has a broken microphone – it's not picking up my voice at all. This is a critical defect for a gaming product. Second, I noticed a duplicate charge for my 'SimpleBest Premium Membership' on my last statement. I was charged twice this month! Please fix the headset and refund the duplicate membership charge immediately. This is quite disappointing.

Sample Message 6: Return/Refund Request (Wrong Item Received)

Subject: Incorrect item sent - Order #CLOTH3456

Hi,
I placed order #CLOTH3456 for a "Blue Summer Dress, size Medium." However, what I received today was a "Red Winter Jacket, size Large." This is completely the wrong item and not what I ordered! I need to return this jacket and either get the correct dress sent to me or receive a full refund. Please provide instructions on how to proceed.

Submitting Your Assessment

- Select the Confirm completion button to submit your assessment. This button is in the browser tab where you initially accessed the assessment.
- Please wait for the validation process to complete, this can take up to 15 minutes.
- Results will be displayed on learning.sap.com under My Certifications.